

SAMYA

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PROFESSIONAL SUMMARY

AI/ML Engineer and MSc Data Science & Artificial Intelligence candidate at Newcastle University with end-to-end expertise in deep learning, NLP, computer vision, and MLOps. Delivered production-grade projects spanning Generative AI (RAG-based systems), document intelligence, fine-tuned transformer models, explainable AI, and deployed ML pipelines via FastAPI and Docker. Proficient in Python, PyTorch, TensorFlow, Hugging Face, LangChain, and cloud-adjacent tooling (AWS, Docker, MLflow). Combines rare cross-disciplinary depth in ML engineering, full-stack development, and data analytics — well-suited to innovation-led teams at organisations including Google, Microsoft, NVIDIA, Apple, and Accenture.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, HTML5, CSS3

ML / DL Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, OpenCV, Hugging Face Transformers, LangChain

Generative AI LLMs: RAG Systems, Prompt Engineering, LLM Integration (Mistral, Hugging Face), Text Generation

AI Specialisms: Deep Learning, CNNs, NLP, Fine-Tuning (BERT), Explainable AI (LIME), Computer Vision, Feature Engineering, Hyperparameter Optimisation, Model Evaluation

MLOps & Cloud: Docker, FastAPI, MLflow, REST APIs, AWS (fundamentals), Git, GitHub, Kubernetes (basics), Streamlit, Hugging Face Spaces, Google Colab Pro, Render

Data & Analytics: Pandas, NumPy, Matplotlib, Seaborn, Power BI (DAX, Power Query), FAISS, EDA, Statistical Reasoning

Web / Full-Stack: React, Node.js, Express.js, REST APIs

Soft Skills: Analytical Thinking, Research & Experimentation, Technical Writing, Cross-functional Collaboration

EDUCATION

Newcastle University

MSc Data Science & Artificial Intelligence

Newcastle upon Tyne, UK

Sep 2025 – Sep 2026 (In Progress)

- Core modules: Machine Learning, Deep Learning, Statistics for Data Science, Applied AI Systems, Data Visualisation, Research Methods.
- Dissertation: Applied ML / Computer Vision (in progress). Tools: Python, TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, Google Colab Pro.

Sharda University

Bachelor of Computer Applications

India

Jul 2021 – Jun 2024

- CGPA: 8.1 / 10. Core study: AI fundamentals, databases, software engineering, data structures, and applied computing.

AI / ML PROJECTS

RAG-Powered Document Q&A Chatbot | *LangChain, Mistral-7B (Hugging Face), FAISS, Streamlit, Python*

- Architected a production-grade **Retrieval-Augmented Generation (RAG)** system enabling semantic question-answering over user-uploaded PDF documents via LangChain pipelines and **FAISS vector search**.
- Integrated Generative AI (Mistral-7B LLM) via Hugging Face Inference API to generate context-aware responses, substantially **reducing LLM hallucinations** through chunk-level retrieval, text splitting, and prompt engineering.
- Deployed a live Streamlit UI on **Hugging Face Spaces**; system directly replicates enterprise document intelligence and knowledge management workflows used at scale in industry.

Fake News Detector with Explainable AI | *BERT, Hugging Face Transformers, LIME, Streamlit, Python*

- Fine-tuned BERT** (bert-base-uncased) on the LIAR dataset for binary misinformation classification; achieved strong accuracy with calibrated confidence scores on held-out evaluation sets.
- Integrated **LIME (Local Interpretable Model-agnostic Explanations)** to generate token-level feature attributions, aligning outputs with AI transparency and regulatory explainability requirements in high-stakes environments.
- Deployed a full Streamlit inference interface on Hugging Face Spaces — demonstrating end-to-end NLP pipeline ownership from data preprocessing through model fine-tuning, interpretability, and live deployment.

Customer Churn Prediction System | *Python, XGBoost, Random Forest, FastAPI, Docker, Streamlit*

- Built a complete supervised ML pipeline — data ingestion, EDA, feature engineering, model training (**Logistic Regression, Random Forest, XGBoost**) — with hyperparameter tuning via stratified cross-validation.
- Selected optimal model using **ROC-AUC, F1-score, and precision-recall** tradeoff analysis; serialised and served predictions via a **FastAPI** REST endpoint containerised with **Docker** and deployed on Render.
- Demonstrates full MLOps discipline: reproducible codebase, CI-ready pipeline structure, and live API inference — directly applicable to production ML engineering roles.

AI Resume Analyser & Job Matcher | *SpaCy, BERT Sentence Embeddings, Hugging Face, Streamlit, Python*

- Engineered an NLP pipeline using **SpaCy NER** for skills extraction from resumes and job descriptions, combined with **sentence-transformer (BERT) embeddings** for semantic similarity scoring.
- Generated match scores and LLM-powered Generative AI recommendations for resume improvement; deployed on **Hugging Face Spaces** with a real-time Streamlit UI serving live inference.
- Demonstrates applied semantic search, embedding-based retrieval, and real-world NLP product development — directly relevant to talent intelligence and AI-powered SaaS applications.

AI-Powered Business Intelligence Platform | *Power BI, DAX, Python, React, Node.js, REST API, FastAPI*

- Designed and built a full-stack **business intelligence platform** combining an interactive **Power BI dashboard** (DAX measures, Power Query transformations) with a Python **FastAPI** backend serving Integrated Generative AI (LLM-based insights) to produce natural language summaries, anomaly detection explanations, and business recommendations.
- Computed revenue, profit margin, customer segmentation, and YoY growth trends using **DAX**; integrated a React frontend to surface AI-driven anomaly alerts and natural language summaries via LLM API calls.
- Merged data analytics, ML inference, and full-stack engineering into a single cohesive product — demonstrating the cross-disciplinary capability required for senior AI/data engineering roles at MNCs.

PROFESSIONAL EXPERIENCE

Software Developer Intern **Jan 2024 – Apr 2024**
Remote
Bhramaand

- Developed full-stack web features using **React, Node.js, and SQL**; built REST APIs, authentication flows, and integrated frontend components with backend services.
- Applied clean coding practices and version control (Git/GitHub) in a structured team environment — demonstrating software engineering discipline relevant to ML pipeline and MLOps work.

Branding & Listing Designer **Apr 2024 – Aug 2025**
Remote
Seller Centric

- Contributed to AI-assisted design workflow optimisation; managed cross-functional documentation and platform compliance under deadline-driven conditions with multiple concurrent deliverables.

Marketing Intern **Jun 2024 – Mar 2025**
Remote
TeamLease RegTech

- Delivered structured content workflows, cross-team coordination, and digital channel outputs with consistent professional execution across a regulated technology environment.

CERTIFICATIONS

- AI/ML MasterClass
- Full Stack Web Development Bootcamp
- HTML, CSS, React & JavaScript Course
- UI/UX Design with Figma

VOLUNTEER & LEADERSHIP

Club Coordinator | GitHub Club, Sharda University **2022–23**
• Organised technical events, online sessions, and community outreach; supported members with coding tools and design.

Event Graphic Lead | Hackathon 2023, Sharda University **2023**
• Led all visual branding: brochures, banners, ID cards, and certificates for the university’s flagship hackathon.

Design Lead | TechNova Technical Society **2022–23**
• Managed event branding and all promotional materials for major technical competitions.

ADDITIONAL INFORMATION

Languages: English (Professional), Hindi (Native)
Availability: Up to 20 hrs/week (open to full-time from Aug 2026)
Location: Newcastle Upon Tyne, UK (Open to Relocate)
Portfolio: behance.net/samya **GitHub:** github.com/samya